

SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND MAINTENANCE

TOPIC 1: Introduction to Solar Photovoltaic (PV) System

Solar energy is energy from the sun. A Photovoltaic (PV) system converts sunlight directly into electricity using solar panels. It is important because it is renewable, clean, reduces electricity bills and is useful in areas without reliable grid power. Components of a basic PV system include: solar panels, charge controller, battery, inverter and cables. Advantages: free fuel (sunlight), low running cost, environmentally friendly, suitable for homes and schools.

TOPIC 2: Electrical Appliances Power Load / Load Sizing

Load means the total electrical power required by all appliances that will be connected to the solar system. To size a system: list each appliance, note its wattage (W), estimate hours of use per day, then calculate Watt-hours ($W \times \text{hours}$). Total daily energy need guides the choice of panel, battery and inverter sizes.

TOPIC 3: Electrical Safety Precautions

Electrical safety means protecting people and property from electric shock, fire and equipment damage. Causes of accidents: faulty wiring, wet hands, overloaded sockets, damaged cables. Basic rules: switch off power before work, use insulated tools, never touch live wires, keep water away from electrical equipment. First aid for electric shock: switch off power source, do not touch victim with bare hands, call for help, check breathing and start CPR if trained.

TOPIC 4: Safety in Handling Solar PV Materials

Handle panels carefully to avoid cracking; store batteries upright in a cool, dry place; use correct cable sizes. Wear safety kits: gloves, goggles, safety boots. Dispose of old batteries properly – they contain harmful chemicals.

TOPIC 5: Solar Panels

Solar panels (modules) convert sunlight into DC electricity. Types: Monocrystalline (high efficiency, dark cells), Polycrystalline (blue cells, slightly lower efficiency), Thin-film (flexible, lower efficiency). Install facing the sun (in Nigeria, generally facing south) at a suitable tilt. Clean panels regularly with soft cloth and water; avoid scratches.

TOPIC 6: Inverters

An inverter converts DC power from panels/batteries into AC power used by most home appliances. Types include pure sine wave and modified sine wave inverters. Keep inverters in a dry, ventilated place and follow manufacturer's maintenance guidelines.

TOPIC 7: Battery Bank Storage

Batteries store energy for use when the sun is not shining. Common types: Lead-acid (cheaper, heavier) and Lithium-ion (lighter, longer life, more expensive). Capacity is measured in Ampere-hours (Ah). Maintain correct electrolyte level (for lead-acid), keep terminals clean, and avoid deep discharge.

TOPIC 8: System Sizing (Battery & Inverter)

System sizing means calculating the correct sizes of panels, battery and inverter to match the load. Match inverter capacity (in Watts or kVA) to the total power of appliances that may run at the same time. Size batteries to store enough energy for the required autonomy (e.g. one night or more).

Summary

Solar PV systems convert sunlight into electricity. Key components are panels, charge controller, battery and inverter. Safety, correct load sizing and proper maintenance are essential for a reliable system.

End of Solar PV First Term Notes